

DAILY ENGINEERING REPORT

End-of-Day Report

Cable BOQ — review & corrections

P-379 / SABI · Tuesday, 17 June 2026

DEVELOPER

Engineering — DrawToBOQ Estimation Pipeline

PROJECT

P-379 / SABI — B+G+8+R Commercial & Residential Tower

FOCUS TODAY

Cable BOQ review against Mr. B.K. Murali's feedback

3

Findings verified

2

Fixed in production

1

Needs drawing re-scan

1

SUMMARY OF TODAY

Today I worked through Mr. B.K. Murali's technical feedback on the P-379 power cable take-off. The feedback flagged that the cable quantities and the lighting schedule did not look like a proper floor-by-floor, project-specific take-off, and questioned whether figures had been carried over from another project.

I confirmed which file he had actually reviewed, opened it, and checked his points line by line against the real numbers. All of his findings hold up — and the issues he raised are already corrected in our live production code. The file he saw was an older sample generated before those fixes landed.

HEADLINE

Murali is right. The main bug — SMDB-to-DB floor boards counted as 1 instead of multiplied per floor — is real in the reviewed sample, but is already fixed in the live system. The reviewed file was just a stale, pre-fix sample.

Status at a glance

WORK ITEM	STATUS
Confirm which file was reviewed	Done
Open file & verify all 3 findings against real data	Done
Map findings to existing production fixes	Done
Answer "will regenerating help?"	Done
Regenerate live P-379 bid through production route	Pending
Re-scan drawings for project-specific lighting	Pending
Port offline sample script (.mjs) to match production	Pending

2 CONTEXT — THE TAKE-OFF METHODOLOGY

For reference, the correct cable take-off procedure from Murali's review has three parts:

Part A · Mains: LV Panel to SMDB

- Start at the Ground Floor LV Panel and open the schematic / single-line drawing.
- List every SMDB the panel feeds, with its reference number (typically one SMDB per floor).
- Against each SMDB record both the cable length and the cable size — size taken from the schematic as a short label (e.g. "150 sq.mm").
- Measure the ground-floor run, then add the floor height (about 5 m) for each floor climbed up the riser.

Part B · Distribution: SMDB to DB

- Each SMDB feeds DBs on its floor. Typical floors repeat, so one floor is calculated and multiplied across all typical floors (e.g. $\times 8$ or $\times 16$).
- A floor-wise schedule must be prepared — measure floor by floor, then total.

Part C · Lighting Fixtures

- Fixture types and naming must be read from this project's own drawing legend — never reused from another or a domestic project.
- Quantities must be broken up floor by floor and then totalled.

3 WHAT I DID — INVESTIGATION

Rather than reason from the code alone, I opened the exact sample file Murali had graded and pulled out its key bills to check his claims against real numbers:

- Opened docs/p379-power-boq-industry.xlsx (file date 15 June 2026 — i.e. before the corrections).
- Confirmed the building from the sheets: B + G + 8 + R (Basement, Ground, 8 typical floors, Roof).
- Inspected Bill 3 (SMDBs), Bill 4 (Distribution Boards), Bill 5 (LV Cables), Bill 8 (Lighting Fixtures).
- Used a one-off ExcelJS dump script to read the cells, then removed it afterwards — no repo changes left behind.

4 WHAT I FOUND — EVIDENCE

CLOSE Finding A · Mains cable (LV to SMDB)

Bill 5 §5.1 lists the main feeders. The per-floor build-up is present — length grows by about 4 m for each floor climbed — which is why this part was "almost matching":

FEEDER (TO FLOOR)	CABLE SIZE	LENGTH (M)
SMDB-1F	4C × 150 mm ²	25
SMDB-2F	4C × 150 mm ²	29
SMDB-3F	4C × 150 mm ²	33
SMDB-4F	4C × 150 mm ²	37
SMDB-5F ... 8F	4C × 150 mm ²	41 / 45 / 49 / 53

Issue: the cable size is buried inside a long, repeated specification string — “4C × 150 mm² XLPE / SWA / PVC 600/1000 V armoured ...” with a generic brand list (Ducab / NCC / Oman / Riyadh). There is no clean, separate cable-size field — the generic, non-project-specific description Murali noted.

ERROR

Finding B · Distribution cable (SMDB to DB) — the principal error

In Bill 4, repeating floor boards were entered as a single line at quantity 1, with no multiplication across floors:

ITEM	DESCRIPTION	QTY SHOWN	QTY EXPECTED
4.1.15	DB-T01 to DB-T15 (odd floors)	1	15 × 4 = 60
4.1.16	DB-T01 to DB-T15 (even floors)	1	15 × 4 = 60
4.1.17	DB-SHOP01 to DB-SHOP12	1	12

So roughly 120 typical-floor DBs were counted as 2 boards, and 12 shop DBs as 1. The matching SMDB-to-DB cables in Bill 5 (§5.3 + §5.4) total only about 21 runs for the whole building, where roughly 120 are expected. The distribution cable quantity is heavily understated — exactly the “multiplication not done” issue raised in the review.

GAP

Finding C · Lighting fixtures — no floor-wise breakdown

Bill 8 lists each fixture type as a single lump quantity for the whole tower, referenced to a drawing range (“P-103 ... P-108”) with generic catalogue naming — no per-floor schedule and no fixture-type tags from this project’s legend:

ITEM	FIXTURE TYPE	QTY (WHOLE BUILDING)
8.1.1	Recessed LED downlight, 12 W	480
8.1.2	Surface LED panel, 600 × 600	60
8.1.3	Linear LED batten, 1200 mm	220
8.1.4	Bulkhead LED, IP65	90
8.1.6	Pendant decorative (provisional)	8

There is no “1F = x, 2F = y ... total” breakdown, confirming the lighting concern.

5

CODE STATUS — ALREADY IN PRODUCTION

I traced each finding to the code. All three are already handled in the live estimation path and run automatically on every production generation:

5.1 Dedicated cable-size column

- Bill 5 now uses a 9-column layout with a separate “Cable Size” column; the spec text prints once per section, not per line.
- Code: `src/lib/excel/dubai-industry-boq-xlsx.ts` (`buildBill5`).

5.2 Floor / range expansion — the main fix

- “DB-T01 to DB-T15 (odd floors)” is auto-expanded into one line per (floor, board). For P-379 that turns ~2 summary lines into ~120 individual runs.
- Floor sets (odd / even / typical / 1F–8F) and tag ranges are both recognised.
- Code: `expandAggregatedCables()` in `src/lib/electrical/derive-cable-paths.ts`.

5.3 Per-floor lighting from the drawing

- A lighting-fixtures extraction step reads the drawing legend per floor; Bill 8 groups fixtures floor by floor, flagging provisional items.
- If the legend was not captured, it falls back to a generic list — so a fresh scan is needed in that case.
- Code: buildBill8() + the lighting_fixtures extraction (prompt + schema).

5.4 Duplicate typical-floor SMDB removal

- Where per-floor boards and a generic typical-floor stack both exist, the duplicate stack and its feeders are removed to avoid double counting.
- Code: dedupeTypicalSmdb() in src/lib/electrical/derive-cable-paths.ts.

5.5 Cable length grounded in floor height

- Feeder length = ~4 m lead-in + (floor index × typical floor height) + 0.5 m termination — matches the review's per-floor height rule using the real floor height, not a flat 5 m.
- Code: gateway-prompts/DRAWTOBOQ_ELECTRICAL_EXTRACT.md (Step 10).

All of the above run via enrichElectricalResult() on the estimate, power-BOQ, and gate routes. A verification script (scripts/verify-p379-improvements.ts) drives the production code over the P-379 data and checks the results.

6 WHY THE REVIEWED FILE STILL SHOWED IT

The BOQ generator exists in two parallel copies:

COPY	LOCATION	HAS THE FIXES?
Production	src/lib/excel/*.ts (live app)	Yes — calls enrichElectricalResult()
Offline sample	scripts/lib/*.mjs (docs sample)	No — never updated; embedded fixture

The reviewed file (15 June) was produced by the offline copy, before the corrections — which is why every issue was genuinely present. A supporting change (src/lib/excel/bill-columns.ts) makes the pricing / brand / reference post-processors read columns by header name, so Bill 5's added column does not corrupt them.

7 WILL REGENERATING IMPROVE IT?

ISSUE	FIXED BY REGEN?	WHY
Mains description (A)	Yes	Presentation fix in buildBill5 — re-renders cleanly.
SMDB-to-DB quantity (B)	Yes	expandAggregatedCables() expands ranges on generation.
Lighting floor-wise (C)	Partial	Only if legend was read; otherwise generic list — re-scan needed.
Values truly from drawing	No	Re-renders stored data; does not re-read drawings.
Old offline sample script	No	That copy is not updated yet.

IN SHORT

Regenerating a live bid through the production route corrects Findings A and B immediately. Lighting and confirmation of project-specific data require a fresh scan of the drawings — not just a re-generation.

8 NEXT STEPS — TOMORROW

- Regenerate the P-379 BOQ through the live production route and confirm the mains description and per-floor distribution quantities come out corrected.
- Trace the stored P-379 extraction to check whether per-floor lighting was captured; if not, queue a drawing re-scan.
- Port the offline sample script (scripts/lib/*.mjs) to match production so the docs sample stops showing stale numbers.
- Flag to the team: confirm with the original preparer whether figures were project-specific or adapted from another project.

9 NOTES & RISKS

- Regeneration only re-renders stored data — it does not re-read drawings. Genuinely wrong source values need a re-scan.
- Lighting still falls back to a generic list when the drawing legend was not captured — the weakest remaining area.
- No production code was changed today; this was investigation, verification, and reporting only.

WRAP-UP

Net for the day — Murali's review is valid and verified against the actual file. The principal quantity error and the cable-size presentation are already fixed in the live system; the reviewed file was simply a pre-fix sample. Remaining work: regenerate the live bid, re-scan for project-specific lighting, and port the stale offline script. No blockers.